Risk Model and Bayesian Shrinkage

DDA3600: Factor Investing

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Fall 2025



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Motivation

- Risk models, which utilize factor models, are another crucial component of active investment management.

- Factor models allow dimensionality reduction, facilitating the computation of the covariance matrix of stocks:

$$\boldsymbol{\Sigma} = \boldsymbol{\beta}\boldsymbol{\Sigma}_f\boldsymbol{\beta}^\top + \boldsymbol{\Sigma}_e,$$

  where $N \times K$ matrix $\boldsymbol{\beta} = [\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \cdots, \boldsymbol{\beta}_N]^\top$ is the factor exposure matrix; $N \times N$ matrix $\boldsymbol{\Sigma}$, $K \times K$ matrix $\boldsymbol{\Sigma}_f$, and $N \times N$ matrix $\boldsymbol{\Sigma}_e$ are the covariance matrices of stocks, factors, and idiosyncratic disturbances, respectively.

- Barra factor model provides good *ex ante* estimate of $\boldsymbol{\Sigma}$. To do so, we need to first estimate factor returns, and then estimate factor covariance matrix using time series of factor returns.

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- The model includes a country factor, P industry factors (denoted as $I_1, \ldots, I_P$), and Q style factors (denoted as $S_1, \ldots, S_Q$). Let $K = 1 + P + Q$, representing a total of K factors:

$$\begin{bmatrix} R^e_{1,t} \\ R^e_{2,t} \\ \vdots \\ R^e_{N,t} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} f_{C,t} + \begin{bmatrix} \beta^{I_1}_{1,t-1} \\ \beta^{I_1}_{2,t-1} \\ \vdots \\ \beta^{I_1}_{N,t-1} \end{bmatrix} f_{I_1,t} + \cdots + \begin{bmatrix} \beta^{I_P}_{1,t-1} \\ \beta^{I_P}_{2,t-1} \\ \vdots \\ \beta^{I_P}_{N,t-1} \end{bmatrix} f_{I_P,t}$$

$$+ \begin{bmatrix} \beta^{S_1}_{1,t-1} \\ \beta^{S_1}_{2,t-1} \\ \vdots \\ \beta^{S_1}_{N,t-1} \end{bmatrix} f_{S_1,t} + \cdots + \begin{bmatrix} \beta^{S_Q}_{1,t-1} \\ \beta^{S_Q}_{2,t-1} \\ \vdots \\ \beta^{S_Q}_{N,t-1} \end{bmatrix} f_{S_Q,t} + \begin{bmatrix} e_{1,t} \\ e_{2,t} \\ \vdots \\ e_{N,t} \end{bmatrix}.$$

- In this model, all stocks have an exposure of 1 to the country factor (as the intercept term serves as the country factor).

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- Stock's excess returns at time t and their factor exposures at time $t-1$ are known, and the factor excess returns are estimated using **weighted least squares** (WLS).
- **This regression-weighting scheme reflects the empirical observation that the idiosyncratic risk of a stock decreases as the market capitalization of the firm increases**:

$$\mathbf{W} = \begin{bmatrix} \frac{\sqrt{s_1}}{\sum_{i=1}^{N}\sqrt{s_i}} & 0 & \cdots & 0 \\ 0 & \frac{\sqrt{s_2}}{\sum_{i=1}^{N}\sqrt{s_i}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\sqrt{s_N}}{\sum_{i=1}^{N}\sqrt{s_i}} \end{bmatrix},$$

where s_i is the market capitalization of stock i.

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- **Barra model uses firm characteristics (after necessary normalization) directly as factor exposures.** (E.g., the BM value of a stock at time $t-1$ is taken as its raw factor exposure.)
- It standardizes the raw value through two steps, i.e., removing the mean and dividing by the standard deviation in the cross-section:
  1. **The model assumes that the market portfolio should be neutral on any style factor, meaning that the market portfolio has zero exposure to all style factors.** Thus, for any style factor, this process involves subtracting the **market cap-weighted** factor exposure from the raw factor exposure. After removing the mean, the factor exposure satisfies:

$$\sum_{i=1}^{N} s_i \beta_{i,t-1}^{S_q} = 0, \quad q = 1, \ldots, Q.$$

  2. The demeaned value is further divided by the cross-sectional standard deviation.

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A Necessary Constraint

- In the model, there exists **multicollinearity between the country factor and the P industry factors.** The exposure vector for the country factor can be expressed as a linear combination of the industry factor exposure vectors.

- The following constraint on industry factor returns is imposed:

$$s_{I_1} f_{I_1,t} + s_{I_2} f_{I_2,t} + \cdots + s_{I_P} f_{I_P,t} = 0,$$

  where s_{I_p} represents the total market-cap-weighted sum of stocks belonging to industry I_p.

- **This constraint ensures that the weighted sum of industry factor returns is zero.** The underlying intuition is that, we want the movements of the industry factors to reflect only their performance relative to the broader market.

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- The intercept term in the cross-sectional regression model serves as the country factor. **The country factor portfolio is approximately the market portfolio weighted by market capitalization.**

- For market portfolio's excess return $R^e_{M,t}$ (let's derive this):

$$\begin{aligned} R^e_{M,t} &= f_{C,t} + \sum_{p=1}^{P} \beta^{I_p}_{M,t-1} f_{I_p t} + \sum_{q=1}^{Q} \beta^{S_q}_{M,t-1} f_{S_q,t} + e_{M,t} \\ &= f_{C,t} + 0 + 0 + \sum_{i=1}^{N} s_i e_{i,t} \\ &\approx f_{C,t}. \end{aligned}$$

- The term $\sum_{i=1}^{N} s_i e_{i,t}$ represents the weighted average of the idiosyncratic returns of all stocks, and therefore its value is approximately zero. This derivation shows that $R^e_{M,t} \approx f_{C,t}$.

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- Express the model in matrix format (also neglect time subscripts for simplicity):

$$\underbrace{\boldsymbol{R}^e}_{N\times 1} = \underbrace{\boldsymbol{\beta}}_{N\times K}\underbrace{\boldsymbol{f}}_{K\times 1} + \underbrace{\boldsymbol{e}}_{N\times 1}.$$

- $\boldsymbol{\beta}$ is the $N \times K$ factor exposure matrix:

$$\beta = \begin{bmatrix} 1 & \beta_1^{I_1} & \cdots & \beta_1^{I_P} & \beta_1^{S_1} & \cdots & \beta_1^{S_Q} \\ 1 & \beta_2^{I_1} & \cdots & \beta_2^{I_P} & \beta_2^{S_1} & \cdots & \beta_2^{S_Q} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 1 & \beta_N^{I_1} & \cdots & \beta_N^{I_P} & \beta_N^{S_1} & \cdots & \beta_N^{S_Q} \end{bmatrix}.$$

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- To incorporate the constraint on industry factor returns, we construct a constraint matrix $\mathbf{C}$. Since there is only one constraint, the relationship among the K factor returns can be expressed as follows:

$$\begin{bmatrix} f_C \\ f_{I_1} \\ \vdots \\ f_{I_P} \\ f_{S_1} \\ \vdots \\ f_{S_Q} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & -\frac{s_{I_1}}{s_{I_P}} & -\frac{s_{I_2}}{s_{I_P}} & \cdots & -\frac{s_{I_{P-1}}}{s_{I_P}} & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} f_C \\ f_{I_1} \\ \vdots \\ f_{I_{P-1}} \\ f_{S_1} \\ \vdots \\ f_{S_Q} \end{bmatrix}.$$

- The matrix on the RHS is the constraint matrix $\mathbf{C}$, which is a $K \times (K-1)$ matrix because there is one constraint among the K factor returns. I pick f_{I_P} to be expressed as a linear combination of the other industry returns. (But you can pick any other industry.)

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- With $\mathbf{W}$ and $\mathbf{C}$ in place, the WLS estimator is:

$$\boldsymbol{f} = \mathbf{C}\left(\mathbf{C}^\top \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} \mathbf{C}\right)^{-1} \mathbf{C}^\top \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{R}^e.$$

- Define a $K \times N$ matrix $\boldsymbol{\Omega}$:

$$\boldsymbol{\Omega} = \mathbf{C}\left(\mathbf{C}^\top \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} \mathbf{C}\right)^{-1} \mathbf{C}^\top \boldsymbol{\beta}^\top \mathbf{W}.$$

- It immediately follows that $\boldsymbol{f} = \boldsymbol{\Omega} \boldsymbol{R}^e$. This means that $\boldsymbol{\Omega}$ represents the factor portfolios for all K factors: the k-th row of $\boldsymbol{\Omega}$ corresponds to the weights of all stocks in the pure factor portfolio for factor k. In other words, the return for each factor at time t is given by:

$$f_{k,t} = \sum_{i=1}^{N} \omega_{k,i} R^e_{i,t}, \quad k = 1, \cdots, K,$$

where $\omega_{k,i}$ is the (k, i) element of the matrix $\boldsymbol{\Omega}$.

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- By multiplying $\mathbf{\Omega}$ with the stock factor exposure matrix $\boldsymbol{\beta}$, we obtain a new matrix $\mathbf{\Omega}\boldsymbol{\beta}$.
- Since $\mathbf{\Omega}$ is a $K \times N$ matrix and $\boldsymbol{\beta}$ is an $N \times K$ matrix, $\mathbf{\Omega}\boldsymbol{\beta}$ is a $K \times K$ matrix, where **the k-th row represents the exposure of factor k's pure factor portfolio on all K factors.**
- Assuming that the K factors are arranged in the order of country factors, P industry factors, and Q style factors, then:

$$\mathbf{\Omega}\boldsymbol{\beta} = \begin{bmatrix} \mathbf{D}_{(1+P)\times(1+P)} & \mathbf{0}_{(1+P)\times Q} \\ \mathbf{0}_{Q\times(1+P)} & \mathbf{I}_{Q\times Q} \end{bmatrix},$$

  where the subscripts of the submatrices represent their dimensions.

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$$\boldsymbol{\Omega\beta} = \begin{bmatrix} \mathbf{D}_{(1+P)\times(1+P)} & \mathbf{0}_{(1+P)\times Q} \\ \mathbf{0}_{Q\times(1+P)} & \mathbf{I}_{Q\times Q} \end{bmatrix},$$

- $\mathbf{0}_{(1+P)\times Q}$ in the upper right corner indicates that the pure factor portfolios for the country and P industry factors have zero exposure to all Q style factors.
- $\mathbf{0}_{Q\times(1+P)}$ in the lower left corner indicates that the pure factor portfolios for the Q style factors have zero exposure to the country and industry factors.
- $\mathbf{I}_{Q\times Q}$ in the lower right corner indicates that each style factor's pure factor portfolio has a unit exposure to itself while having zero exposure to all other style factors.

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- $\mathbf{I}_{Q\times Q}$ in the lower right corner indicates that each style factor's pure factor portfolio has a unit exposure to itself while having zero exposure to all other style factors.

A style factor's pure factor portfolio has no exposure to all factors other than itself.

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- Let's consider $\mathbf{D}_{(1+P)\times(1+P)}$ through an example:

$$\begin{bmatrix} 1 & 0.03 & 0.02 & 0.06 & 0.01 & \cdots & 0.02 & 0.01 & 0.03 & 0.05 \\ 0 & 0.97 & -0.02 & -0.06 & -0.01 & \cdots & -0.02 & -0.01 & -0.03 & -0.05 \\ 0 & -0.03 & 0.98 & -0.06 & -0.01 & \cdots & -0.02 & -0.01 & -0.03 & -0.05 \\ 0 & -0.03 & -0.02 & 0.94 & -0.01 & \cdots & -0.02 & -0.01 & -0.03 & -0.05 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & -0.03 & -0.02 & -0.06 & -0.01 & \cdots & -0.02 & 0.99 & -0.03 & -0.05 \\ 0 & -0.03 & -0.02 & -0.06 & -0.01 & \cdots & -0.02 & -0.01 & 0.97 & -0.05 \\ 0 & -0.03 & -0.02 & -0.06 & -0.01 & \cdots & -0.02 & -0.01 & -0.03 & 0.95 \end{bmatrix}.$$

- The first row represents the exposure of the country factor portfolio to itself and the industry factors. It is clear and intuitive that the country factor has an exposure of 1.0 to itself. Additionally, the country factor portfolio also has a positive exposure to each industry.

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- Now, let's take a look at the industry factors. For example, the second row represents the exposure of the pure factor portfolio for industry 1 to both the country and industry factors.

- To explain this, consider the following long/short hedged portfolio:
  - long leg goes long all stocks belonging to industry 1. Its factor exposure is:

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \end{bmatrix}.$$

  - short leg is the country factor portfolio, which has the following exposure:

$$\begin{bmatrix} 1 & 0.03 & 0.02 & 0.06 & 0.01 & \cdots & 0.02 & 0.01 & 0.03 & 0.05 \end{bmatrix}.$$

- The industry factor portfolio:

$$\begin{bmatrix} 0 & 0.97 & -0.02 & -0.06 & -0.01 & \cdots & -0.02 & -0.01 & -0.03 & -0.05 \end{bmatrix}.$$

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Pure Factor Portfolios – Industry Factor

- The industry pure factor portfolio is to be 100% long in that industry while simultaneously being 100% short in the country factor portfolio.
- Since the country factor is approximately equivalent to the market portfolio, this pure factor portfolio has a very intuitive interpretation: **it represents the excess return of the industry relative to the market portfolio.**
- This interpretation aligns with the constraint that the weighted sum of industry factor returns equals zero.

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$$\boldsymbol{\Omega\beta} = \begin{bmatrix} \mathbf{D}_{(1+P)\times(1+P)} & \mathbf{0}_{(1+P)\times Q} \\ \mathbf{0}_{Q\times(1+P)} & \mathbf{I}_{Q\times Q} \end{bmatrix},$$

- The pure factor portfolios for the industry and style factors have zero exposure to the country factor.
- For any factor k other than the country factor, the weight of stock i in its pure factor portfolio is $\omega_{k,i}$.
- Zero exposure to the country factor implies that $\sum_{i=1}^{N} \omega_{k,i} = 0$, meaning the sum of the weights of all stocks in the portfolio is zero, indicating that it is *dollar neutral*.

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- The pure factor portfolio for the **country factor** is an approximate market portfolio, which is a pure long portfolio. This portfolio has an exposure of 1 to the country factor, positive exposures to all industries, and zero exposure to all style factors.

- The pure factor portfolio for an **industry factor** is dollar neutral. For industry I_p, this portfolio is 100% long on that industry and 100% short on the country factor portfolio, reflecting the excess return of the industry relative to the market portfolio, with zero exposure to all style factors.

- The pure factor portfolio for a **style factor** is also dollar neutral. For style factor S_q, its pure factor portfolio has one unit exposure of only to itself, with no exposure to any other industry or style factors. This portfolio's excess returns is solely due to being exposed to that factor.

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- The covariance matrix is a critical input for asset allocation. Accurate estimation is essential for solving the portfolio optimization problem.

- However, the sample covariance matrix is not an ideal estimator, especially when the number of assets N is close to the number of periods T.

- We can use the Frobenius norm to measure the difference between two matrices:

$$||\mathbf{A} - \mathbf{B}||_F = \sqrt{\sum_{i,j} |a_{ij} - b_{ij}|^2}$$

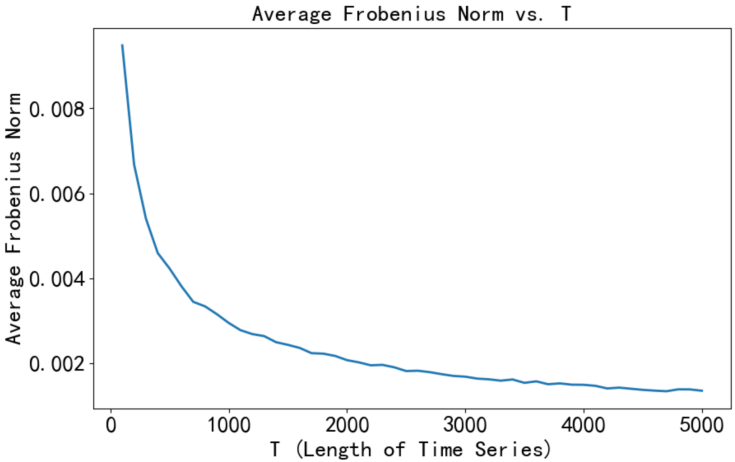
where a_{ij} and b_{ij} are elements in the i-th row and j-th column of matrices $\mathbf{A}$ and $\mathbf{B}$, respectively.

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- Assume a known data-generating process (DGP) to illustrate the relationship between T and N.
- Set $N = 50$; returns follow a multivariate normal distribution with mean zero.
- A randomly generated positive-definite matrix is used as the true covariance matrix.
- Simulation setup:
  - Let T vary from 100 to 5000 in increments of 100.
  - For each T, perform $M = 1000$ simulations.
  - In each simulation, calculate the Frobenius norm between the sample covariance matrix and the true covariance matrix; take the mean over 1000 simulations as the error for that T.

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- For $N = 50$, at least $T \geq 2000$ data points are needed for the sample covariance matrix to be reliable.
- This requires approximately 8-10 years of daily data, assuming 250 trading days per year.
- In real-world asset allocation, the number of assets often exceeds 50. Better covariance matrix estimators are needed. **Solution: Bayesian Shrinkage.**



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- In finance, Bayesian methods are particularly valuable because:
  - Financial data are often noisy and samples can be limited.
  - Relying solely on data may lead to overfitting or unstable estimates.

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- In finance, Bayesian methods are particularly valuable because:
  - Financial data are often noisy and samples can be limited.
  - Relying solely on data may lead to overfitting or unstable estimates.

Incorporating a prior belief can stabilize estimates, mitigating the unreliability of purely data-driven approaches.

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Challenges of Specifying a Prior

- Specifying an appropriate prior can be very challenging.
- Financial data are complex and volatile, making it difficult to define a precise prior distribution.
- **Example:** Estimating a covariance matrix in a high-dimensional portfolio.
  - In theory, we could assign a prior distribution to the covariance matrix, often a Wishart distribution.
  - But in high-dimensional settings:
    - Calculating the posterior is complex and computationally prohibitive.
    - Choosing a specific prior can introduce bias in the absence of strong prior knowledge.
    - Financial data are heterogeneous and noisy, making a fixed prior inappropriate.

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Practitioners often use Bayesian updating without an explicit prior distribution – the Shrinkage approach.

- Apply Bayesian principles—updating beliefs based on data—without defining a complex prior.
- The shrinkage approach starts with a conservative target and adjust the estimate based on sample data.

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- The bias statistic is a commonly used metric for evaluating the accuracy of risk models. It measures the discrepancy between predicted and actual risk values. If there is a significant differnece between them, the risk forecast is considered biased, hence the term *bias statistic*.

- The bias statistic B_i for asset i is defined as:

$$B_i = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}\left(b_{i,t} - \bar{b}_i\right)^2}, \quad b_{i,t} = \frac{R^e_{i,t}}{\sigma_{i,t}},$$

  where $\sigma_{i,t}$ is the volatility forecast, $b_{i,t}$ represents the standardized return, and $\bar{b}_i$ is the average of $b_{i,t}$.

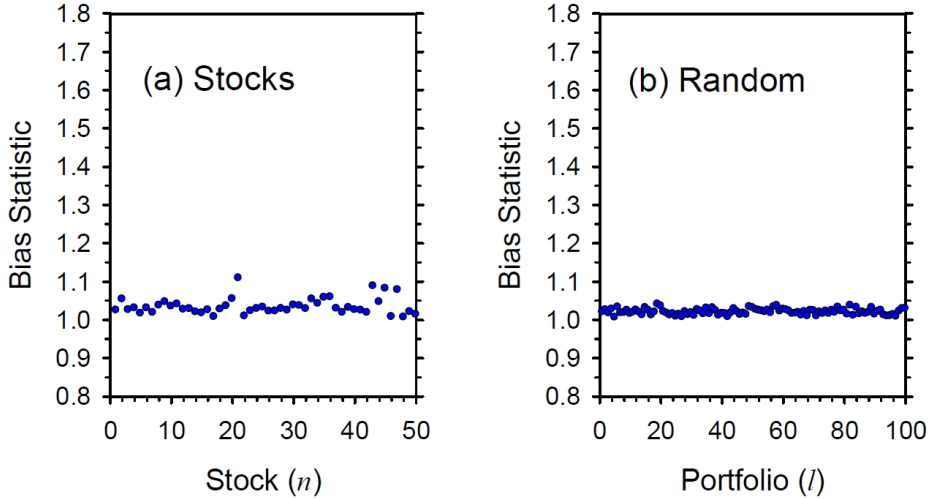
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- If the *ex ante* risk estimate $\sigma_{i,t}$ is accurate, then the standard deviation of the standardized return $b_{i,t}$ should be 1.
- When T is large, B_i approximately follows a normal distribution, with a 95% confidence interval given by:

$$B_i \in \left[1 - \sqrt{2/T}, 1 + \sqrt{2/T}\right].$$

- If B_i falls outside this range, it suggests that the ex ante risk estimate $\sigma_{i,t}$ may be inaccurate.

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- Source: Menchero, Wang and Orr (2011).

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We Do Not Invest Randomly

4 Barra's Eigenfactor Adjustment

- In reality, investments are not random. Stocks are combined to construct **optimal portfolios**.
- The stock covariance matrix $\mathbf{\Sigma}$ is a critical input for constructing portfolios such as:
  - Minimum Variance Portfolio
  - Mean-Variance Efficient Portfolio
- These portfolios optimize based on different objective functions, but both rely on an accurate estimate of $\mathbf{\Sigma}$.

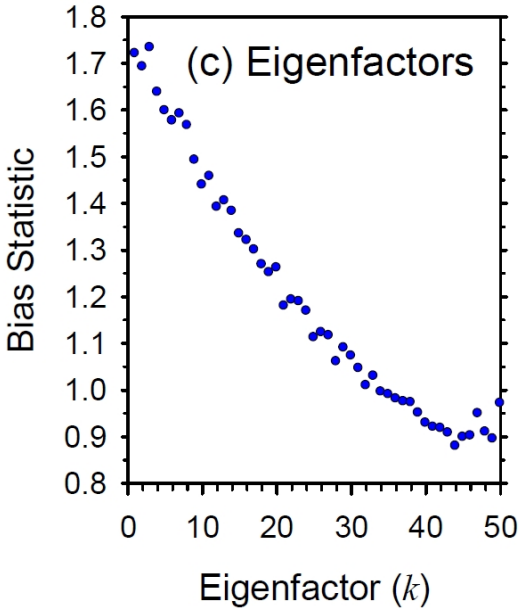
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- These portfolios optimize based on different objective functions, but both rely on an accurate estimate of $\mathbf{\Sigma}$.

So, how does the bias statistic behave for these optimized portfolios?

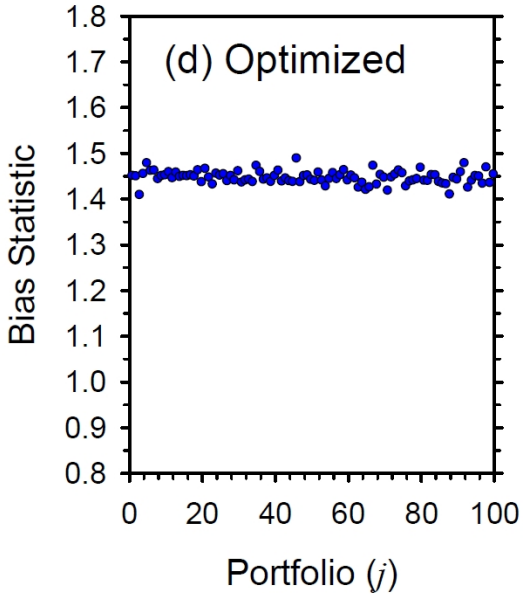
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- To study the impact of $\mathbf{\Sigma}$ on optimized portfolios, we apply eigen-decomposition to the sample covariance matrix $\hat{\mathbf{\Sigma}}$.
- Each eigenvector is an $N \times 1$ vector, where each entry represents the weight of a stock in that eigenfactor portfolio.
- Using these eigenvector values as weights, we create **eigenfactor portfolios**.
- **Eigenfactor portfolios with smaller sample variances tend to have much larger bias statistics.**



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- Do these eigenfactor portfolios matter in practice? They do matter!!
- Eigenfactors play an important role in portfolio optimization.
  - E.g., the eigenfactor with the smallest eigenvalue solves for the minimum variance portfolio subject to the constraint that the sum of squared weights adds up to 1.
- We can expect that the bias statistics for the optimized portfolios derived from them also deviate from 1.0.



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- Wait a minute!!! All the evidence mentioned above are related to sample covariance matrix of **individual stocks**. How does this relate to factor covariance matrix?
- Suppose the optimal weights of stocks is $\boldsymbol{w}$. Then, the variance of this portfolio is $\sigma_p^2 = \boldsymbol{w}^\top \hat{\boldsymbol{\Sigma}} \boldsymbol{w}$.
- Using a factor model, the accuracy of σ_p^2 largely depends on $\boldsymbol{w}^\top \boldsymbol{\beta} \hat{\boldsymbol{\Sigma}}_f \boldsymbol{\beta}^\top \boldsymbol{w}$.
- Define $\boldsymbol{w}_f = (\boldsymbol{w}^\top \boldsymbol{\beta})^\top$, so that $\sigma_p^2 \approx \boldsymbol{w}_f^\top \hat{\boldsymbol{\Sigma}}_f \boldsymbol{w}_f$.

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Individual Stocks vs. Factors

4 Barra's Eigenfactor Adjustment

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- Define $\boldsymbol{w}_f = (\boldsymbol{w}^\top \boldsymbol{\beta})^\top$, so that $\sigma_p^2 \approx \boldsymbol{w}_f^\top \hat{\boldsymbol{\Sigma}}_f \boldsymbol{w}_f$.

The focus on the bias statistic for the optimal stock portfolio can be shifted to the bias statistic for the optimal portfolio of factors.

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Eigenfactor Portfolios of Factors

4 Barra's Eigenfactor Adjustment

- Like the case for individual stocks, we apply eigen-decomposition to the sample **factor** covariance matrix $\hat{\mathbf{\Sigma}}_f$: $\hat{\mathbf{\Sigma}}_f = \mathbf{U}\mathbf{D}\mathbf{U}^\top$, where $\mathbf{U}$ is the eigenvector matrix and $\mathbf{D}$ is a diagonal matrix with eigenvalues on the diagonal.

- Each eigenvector is a portfolio of the K factors, known as an **eigenfactor portfolio**. They form an orthogonal basis.

- When using these K factors as assets, any set of weights $\boldsymbol{w}_f$ can be expressed as a linear combination of this orthogonal basis.

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Eigenfactor Portfolios of Factors

4 Barra's Eigenfactor Adjustment

- Like the case for individual stocks, we apply eigen-decomposition to the sample **factor** covariance matrix $\hat{\mathbf{\Sigma}}_f$: $\hat{\mathbf{\Sigma}}_f = \mathbf{U}\mathbf{D}\mathbf{U}^\top$, where $\mathbf{U}$ is the eigenvector matrix and $\mathbf{D}$ is a diagonal matrix with eigenvalues on the diagonal.

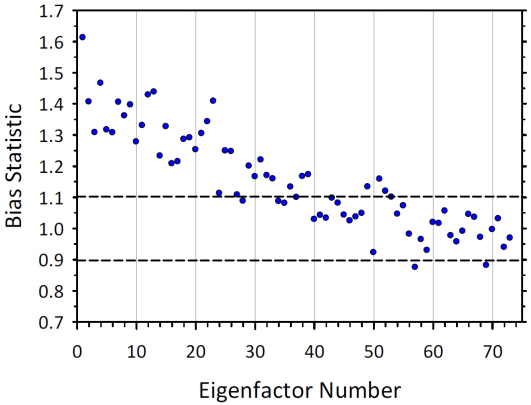
- Each eigenvector is a portfolio of the K factors, known as an **eigenfactor portfolio**. They form an orthogonal basis.

- When using these K factors as assets, any set of weights $\boldsymbol{w}_f$ can be expressed as a linear combination of this orthogonal basis.

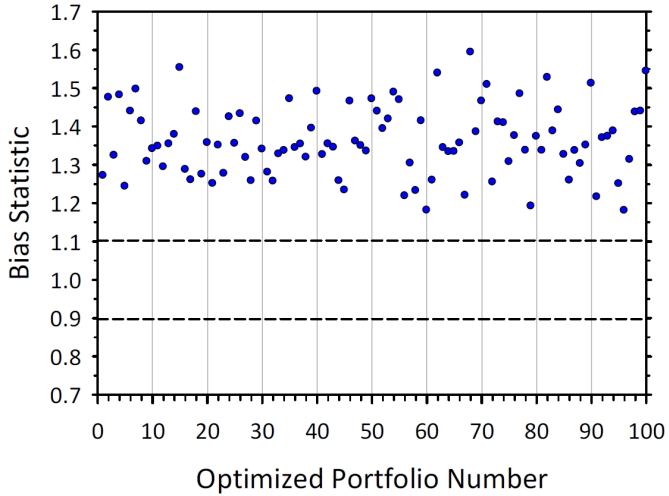
Whether the ex ante estimate $\boldsymbol{w}_f^\top \hat{\mathbf{\Sigma}}_f \boldsymbol{w}_f$ is accurate depends on whether the ex ante variance estimates for these K eigenfactor portfolios are accurate. But...

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- Empirical studies, e.g., USE4 model, show that the bias statistic for these eigenfactor portfolios is not always close to 1.0.
- **The smaller the ex ante variance estimate of the eigenfactor portfolio, the higher its ex post bias statistic, often significantly greater than 1.0.**
- Given the inaccuracy of these estimates, it is unlikely that $\boldsymbol{w}_f^\top \hat{\boldsymbol{\Sigma}}_f \boldsymbol{w}_f$ will be accurate either.



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- Source: Menchero, Orr and Wang (2011).

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Solution: Eigenfactor Adjustment

The core of the eigenfactor adjustment method is to correct for the bias in eigenvalues (i.e., the variances of eigenfactors) via simulation.

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Solution: Eigenfactor Adjustment

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- This approach treats the sample factor covariance matrix $\hat{\mathbf{\Sigma}}_f$ as the "population".
- Through simulation, it generates "sample" covariance matrices, denoted as $\tilde{\mathbf{\Sigma}}_f$. Eigenfactor covariance matrices $\mathbf{D}$ and $\tilde{\mathbf{D}}$ are derived from $\hat{\mathbf{\Sigma}}_f$ and $\tilde{\mathbf{\Sigma}}_f$, respectively.

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- The bias between $\tilde{\mathbf{D}}$ and $\mathbf{D}$ approximates the bias between $\mathbf{D}$ and the true but unknown eigenfactor covariance matrix.

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- The bias between $\tilde{\mathbf{D}}$ and $\mathbf{D}$ approximates the bias between $\mathbf{D}$ and the true but unknown eigenfactor covariance matrix.
- Use this bias to adjust $\mathbf{D}$, and subsequently recalculating $\hat{\mathbf{\Sigma}}_f$. This improves the estimate of the true covariance matrix $\mathbf{\Sigma}_f$.

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- **Step 1:** Perform eigen-decomposition on $\hat{\mathbf{\Sigma}}_f$ to obtain $\mathbf{U}$ and $\mathbf{D}$.

- **Step 2:** Let D_k denote the k-th diagonal element of $\mathbf{D}$. For the eigenfactor portfolio k, assume its returns follow a normal distribution $\mathcal{N}(0, D_k)$. In the m-th simulation, generate a sequence of returns for all T periods for eigenfactor k. Apply this procedure to all K eigenfactors to obtain a $K \times T$ matrix of returns, denoted $\mathbf{b}_m$. Then, use $\mathbf{b}_m$ and $\mathbf{U}$ to compute the factor return matrix $\tilde{\mathbf{f}}_m$:

$$\tilde{\mathbf{f}}_m = \mathbf{U}\mathbf{b}_m.$$

- **Step 3:** Calculate the "sample" covariance matrix of the factor returns $\tilde{\mathbf{f}}_m$:

$$\tilde{\mathbf{\Sigma}}_{f,m} = \mathrm{cov}(\tilde{\mathbf{f}}_m, \tilde{\mathbf{f}}_m).$$

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Procedure, Cont'd

4 Barra's Eigenfactor Adjustment

- **Step 4:** Perform eigen-decomposition on $\tilde{\mathbf{\Sigma}}_{f,m}$ to obtain the corresponding $\tilde{\mathbf{U}}_m$ and $\tilde{\mathbf{D}}_m$.
- **Step 5:** Since the "true" covariance matrix $\hat{\mathbf{\Sigma}}_f$ is known, calculate the "true" eigenfactor covariance matrix using $\hat{\mathbf{\Sigma}}_f$ and $\tilde{\mathbf{U}}_m$:

$$\mathbf{D}_m = \tilde{\mathbf{U}}_m^\top \hat{\mathbf{\Sigma}}_f \tilde{\mathbf{U}}_m.$$

- **Step 6:** Repeat steps 2 to 5 for M simulations. In each simulation, for a given eigenfactor k, calculate the difference between the k-th diagonal element of $\mathbf{D}_m$ and $\tilde{\mathbf{D}}_m$, then take the average of all M simulations to obtain the mean bias:

$$\nu_k = \frac{1}{M} \sum_{m=1}^{M} \sqrt{\frac{D_k^m}{\tilde{D}_k^m}}.$$

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- **Step 7:** The simulations assume normality and stationarity in returns. However, real-world return series do not fully satisfy these assumptions, so additional scaling is applied:

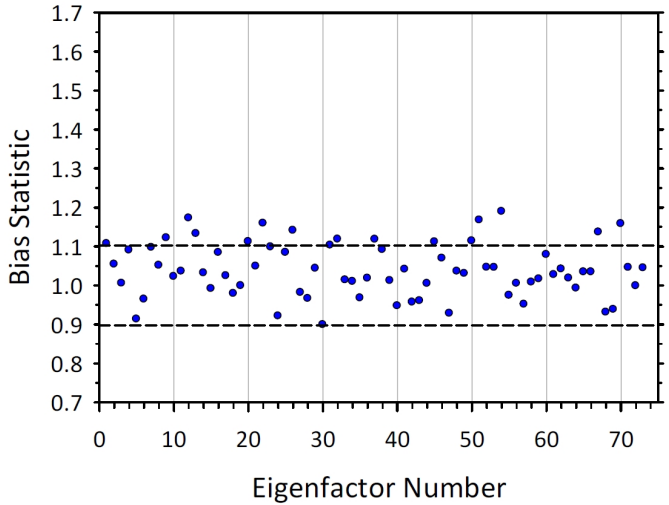
$$\nu_k^s = a(\nu_k - 1) + 1,$$

where a is an empirical coefficient, typically between 1 and 2.

- **Step 8:** ν_k^s is the *correction coefficient* for the variance of eigenfactor k. Use ν_k^s to adjust $\mathbf{D}$ and the factor covariance matrix:

$$\mathbf{D}_0 = \boldsymbol{\nu}^2\mathbf{D}, \quad \hat{\mathbf{\Sigma}}_{f,0} = \mathbf{U}\mathbf{D}_0\mathbf{U}^\top.$$

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- Source: Menchero, Orr and Wang (2011).

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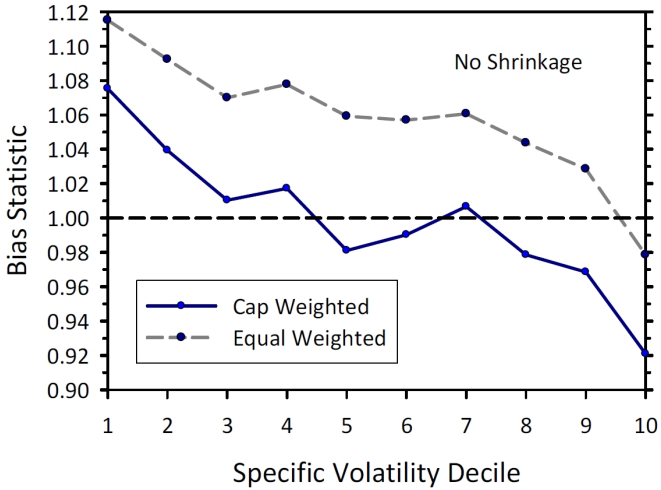
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5 Adjustment of the Idiosyncratic Covariance Matrix

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- Appendix: CNE6 Style Factors

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- Barra also observed that idiosyncratic risk volatility calculated using in-sample data exhibits poor persistence out-of-sample.
- Stocks are divided into deciles based on **idiosyncratic volatility**. Average Bias statistic of each decile is calculated.
- For **low-volatility** deciles, the Bias statistic is greater than 1, indicating an **underestimation**. For **high-volatility** deciles, the Bias statistic is less than 1, suggesting **overestimation**.



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- To address this issue, Bayesian shrinkage is used with a prior for the idiosyncratic volatilities. Barra calculates the prior by averaging the idiosyncratic volatilities of all stocks within the same **size** decile.

- The prior $\bar{\sigma}(\mathcal{S}_i)$ is given by:

$$\bar{\sigma}(\mathcal{S}_i) = \sum_{i \in \mathcal{S}_i} w_i \hat{\sigma}_i,$$

  where w_i is the market cap weight of stock i in decile $\mathcal{S}_i$.

- This weighted average captures the specific risk volatility within a peer group of similar-sized stocks.

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- With the prior $\bar{\sigma}(\mathcal{S}_i)$ and the sample idiosyncratic volatility $\hat{\sigma}_i$, the adjusted idiosyncratic volatility is:

$$\sigma_i^{\mathrm{SH}} = \nu_i \bar{\sigma}(\mathcal{S}_i) + (1 - \nu_i)\hat{\sigma}_i,$$

  where ν_i is the shrinkage intensity, balancing the influence of the prior and the sample.

- A higher ν_i implies greater reliance on the prior, while a lower ν_i suggests confidence in the sample idiosyncratic volatility.

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- The shrinkage intensity coefficient ν_i is defined as:

$$\nu_i = \frac{q|\hat{\sigma}_i - \bar{\sigma}(\mathcal{S}_i)|}{\sqrt{\frac{1}{N(\mathcal{S}_i)} \sum_{j \in \mathcal{S}_i} (\hat{\sigma}_j - \bar{\sigma}(\mathcal{S}_i))^2} + q|\hat{\sigma}_i - \bar{\sigma}(\mathcal{S}_i)|},$$

  where q is an empirical coefficient and $N(\mathcal{S}_i)$ is the number of stocks in size decile $\mathcal{S}_i$.

- The term $|\hat{\sigma}_i - \bar{\sigma}(\mathcal{S}_i)|$ represents the deviation between the sample and prior.

- $\sqrt{\frac{1}{N(\mathcal{S}_i)} \sum_{j \in \mathcal{S}_i} (\hat{\sigma}_j - \bar{\sigma}(\mathcal{S}_i))^2}$ is the standard deviation of the idiosyncratic volatilities of all stocks in $\mathcal{S}_i$, representing the average deviation for this group of stocks.

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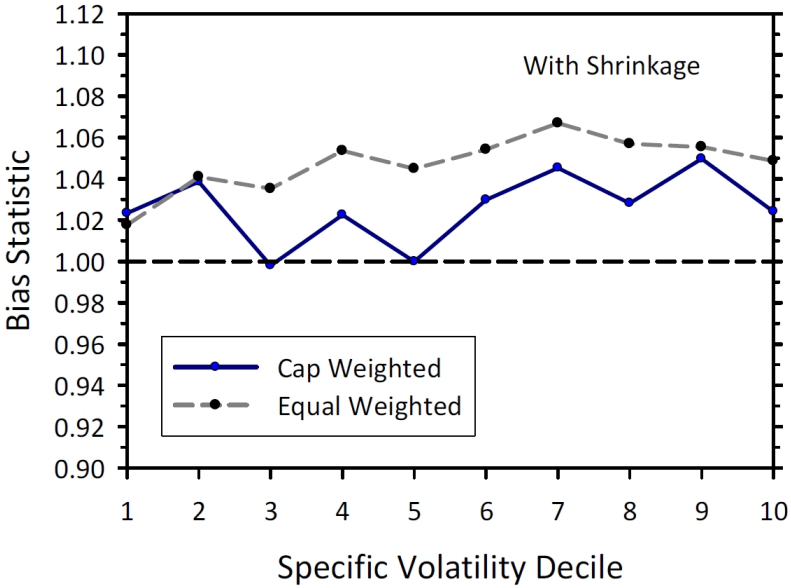


Interpretation of Shrinkage Intensity

- The shrinkage intensity ν_i balances the prior and sample idiosyncratic volatility based on deviation.
- When $|\hat{\sigma}_i - \bar{\sigma}(\mathcal{S}_i)|$ is large, ν_i increases, placing more weight on the prior due to unreliability in the sample data.
- When $|\hat{\sigma}_i - \bar{\sigma}(\mathcal{S}_i)|$ is small, ν_i decreases, reflecting greater confidence in the sample idiosyncratic volatility.
- The empirical coefficient q is chosen to reduce overall Bias statistic across all stocks.

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- Bayesian shrinkage significantly improves out-of-sample performance for idiosyncratic volatility across market cap deciles.



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- Let $\mathbf{X}$ be a $T \times N$ return matrix (assume each column is demeaned). Then the sample covariance matrix is:

$$\mathbf{S}_T = \frac{1}{T-1}\mathbf{X}^\top \mathbf{X}.$$

- The Ledoit-Wolf linear shrinkage estimator is:

$$\hat{\mathbf{\Sigma}}_T = w\mu_T \mathbb{I} + (1-w)\mathbf{S}_T,$$

  where $\mu_T \mathbb{I}$ acts as a conservative target, and w is the shrinkage coefficient balancing the sample data and target.

- μ_T is the mean variance of all assets. The optimal shrinkage weight w is determined by minimizing the mean squared error $\mathbb{E}[||\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}||_F^2]$.

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- **Inspired by factor models in asset pricing,** Ledoit and Wolf also proposed shrinking toward a factor model structure:

$$\boldsymbol{\Sigma} = \boldsymbol{\beta}\boldsymbol{\Sigma}_f\boldsymbol{\beta}^\top + \boldsymbol{\Sigma}_e,$$

  where $\boldsymbol{\beta}$ represents factor loadings, $\boldsymbol{\Sigma}_f$ is the factor covariance, and $\boldsymbol{\Sigma}_e$ is the idiosyncratic variance matrix.

- This captures systematic asset relationships, improving stability by reflecting shared risk factors.

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- Eigenvalue or spectral shrinkage is an extension of the Ledoit-Wolf approach, where we focus on modifying the eigenvalues of the covariance matrix, while keeping the eigenvectors unchanged.
- This method addresses a common issue in high-dimensional covariance estimation where sample eigenvalues tend to be unreliable, with some values inflated by noise and others diminished by sampling error.
- This approach was also applied in Barra's eigenfactor adjustment, where eigenvalues are adjusted to improve stability!

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- Recall that the shrinkage toward the identity matrix is:

$$\hat{\mathbf{\Sigma}}_T = w\mu_T\mathbb{I} + (1-w)\mathbf{S}_T.$$

- This estimator is equivalent to first performing eigen-decomposition on the sample covariance matrix, and then shrinking the eigenvalues:

$$\begin{aligned} \mathbf{S}_T &= \mathbf{U}_T\mathbf{\Delta}_T\mathbf{U}_T^\top, \\ \mathbf{\Delta}_T &\Rightarrow \mathbf{\Delta}_T^\star, \end{aligned}$$

  where $\mathbf{U}_T$ is the matrix of eigenvectors, $\mathbf{U}_T$ is the diagonal matrix of eigenvalues, and $\mathbf{\Delta}_T^\star$ is the shrunk diagonal matrix.

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- Eigenvalue shrinkage adjusts each original eigenvalue λ_i of $\mathbf{S}_T$ toward a target μ_T using the shrinkage coefficient w:

$$\lambda_i^\star = w\mu_T + (1-w)\lambda_i,$$

  where:
  - $\lambda_i^\star$: Shrunk eigenvalue,
  - μ_T: Mean variance of all assets $\equiv$ mean of all eigenvalues.

- **Thus, the logic behind this shrinkage approach is to pull the eigenvalues toward their mean.**

- After shrinking the eigenvalues, we reconstruct the covariance estimator $\hat{\mathbf{\Sigma}}_T$:

$$\hat{\mathbf{\Sigma}}_T = \mathbf{U}_T \mathbf{\Delta}_T^\star \mathbf{U}_T^\top,$$

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Motivation for Nonlinear Shrinkage

- Linear shrinkage applies the same shrinkage factor w to all eigenvalues. However, nonlinear shrinkage allows each eigenvalue to shrink by a different amount, based on data-driven optimization.
- This complex method tailors shrinkage for each eigenvalue, optimizing the diagonal elements of $\mathbf{\Delta}_T^\star$ to further improve the stability of the covariance matrix.
- By focusing on adjusting eigenvalues, we retain essential structure in the covariance matrix while minimizing instability. This approach is commonly used in risk management to improve covariance estimates.
- For more on nonlinear shrinkage, see Ledoit and Wolf (2015, 2020).

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Empirical Bayes Approach

6 Ledoit-Wolf Shrinkage

- In high-dimensional problems, an exact prior distribution may be impractical. The Ledoit-Wolf empirical Bayes approach allows Bayesian updating by applying data-driven shrinkage instead of a fixed prior.

- Factor models and eigenvalue shrinkage improve stability without sacrificing flexibility, adapting to complex data structures.

- To evaluate covariance matrix estimators, construct a **minimum variance portfolio** and compare out-of-sample variances.

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Empirical Bayes Approach

6 Ledoit-Wolf Shrinkage

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- Factor models and eigenvalue shrinkage improve stability without sacrificing flexibility, adapting to complex data structures.
- To evaluate covariance matrix estimators, construct a **minimum variance portfolio** and compare out-of-sample variances.

If an estimator is more accurate, the minimum variance portfolio should exhibit lower out-of-sample volatility.

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- Descriptions of the style factors come from *Barra China A Total Market Equity Model for Long-Term Investors – Empirical Notes* by Jay Yao and Andrei Morozov, issued in August 2018.

- These factors include: Beta, Book-to-Price, Dividend Yield, Earnings Quality, Earnings Variability, Earnings Yield, Growth, Investment Quality, Leverage, Liquidity, Long-Term Reversal, Mid Capitalization, Momentum, Profitability, Residual Volatility, and Size.


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Barra China A Total Market Equity Model for Long-Term Investors

Empirical Notes

Jay Yao

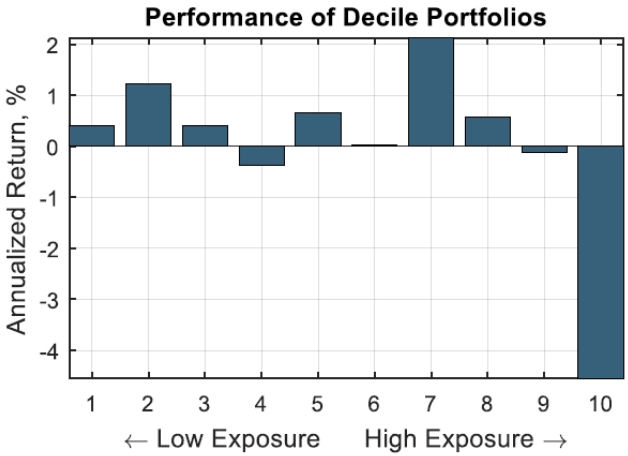
Andrei Morozov

August 2018

AUGUST 2018

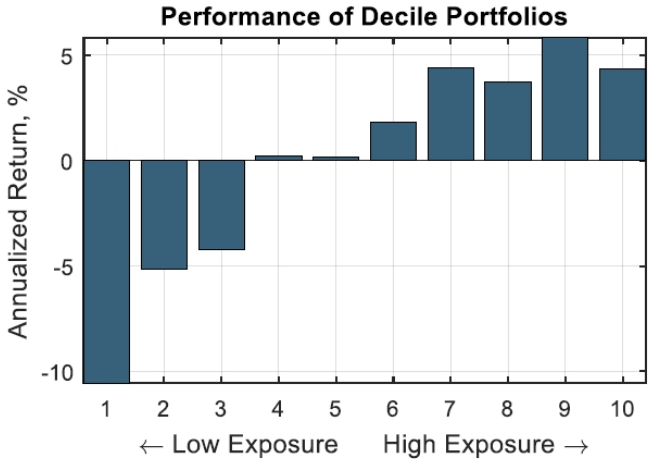
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- **Description:** Captures the market risk that cannot be explained by the Country factor.
- **Motivation:** CAPM.
- **Exposure:** A positive exposure indicates a high beta stock. A negative exposure indicates a low beta stock.
- **Variable:** Historical beta.



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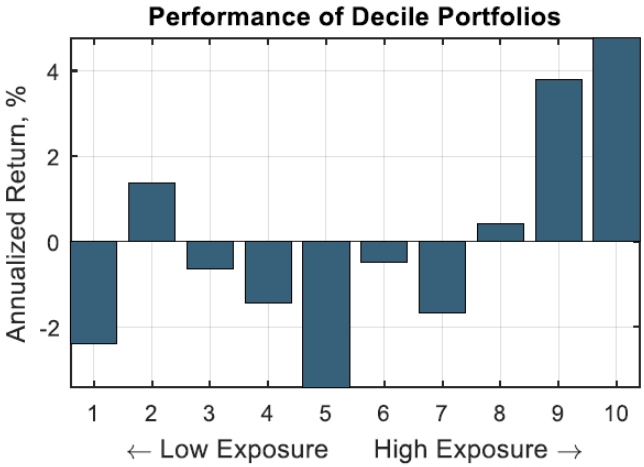
- **Description:** Explains the return component attributable to a stock's book-to-price ratio and can be an indicator of value.
- **Motivation:** Value: high-value companies earned higher returns and experienced higher risk relative to low-value companies.
- **Exposure:** A positive exposure indicates an undervalued stock. A negative exposure indicates an overvalued stock.
- **Variable:** Book-to-Price ratio.



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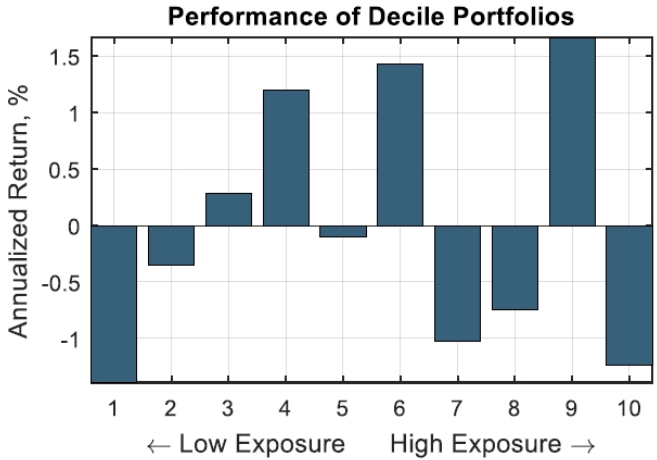
Dividend Yield

- **Description:** Captures differences in stock returns attributable to stock's dividend-to-price ratios.
- **Motivation:** Dividend is one of the central inputs in Gordon Growth Model for valuing a company's stock price.
- **Exposure:** A positive exposure indicates a high dividend yield. A negative exposure indicates a low dividend yield.
- **Variable:** Historical Dividend-to-Price ratio and analyst-predicted Dividend-to-Price ratio.



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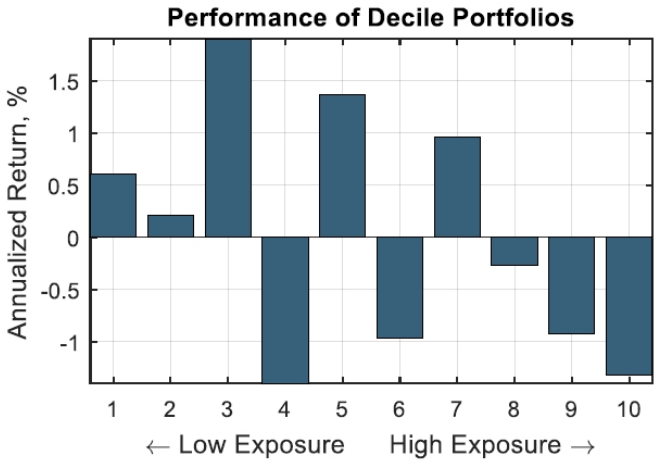
- **Description:** Explains stock return differences due to the accrual components of their earnings.
- **Motivation:** The accrual components of earnings involve significant management discretion and are more prone to manipulations, and thus reflecting low quality.
- **Exposure:** A positive exposure indicates low accruals. A negative exposure indicates high accruals.
- **Variable:** Accruals - Balance Sheet Version and Accruals - Cash flow Version.



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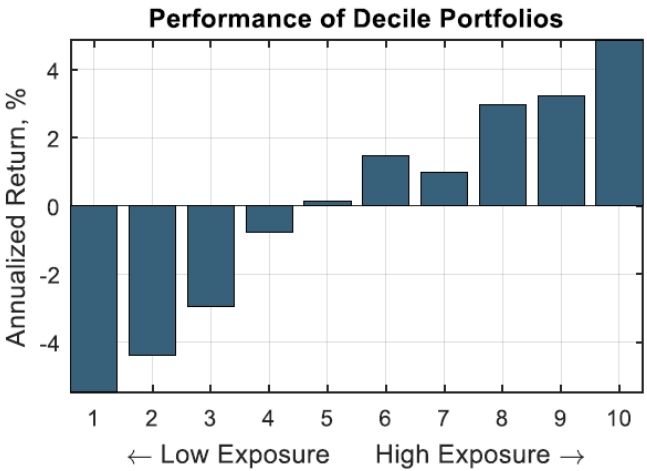
Earnings Variability

- **Description:** Explains stock return differences due to variability in earnings, sales, and cash-flow, and also in analysts' predicted earnings-to-price.
- **Motivation:** Companies with higher earnings quality have less uncertainty around their fundamentals and tend to outperform the market average over the long run.
- **Exposure:** A positive exposure indicates high uncertainty. A negative exposure indicates low uncertainty.
- **Variable:** Std dev of analyst forecast earnings-to-price / variability in earnings / variability in cash flows / variability in sales.



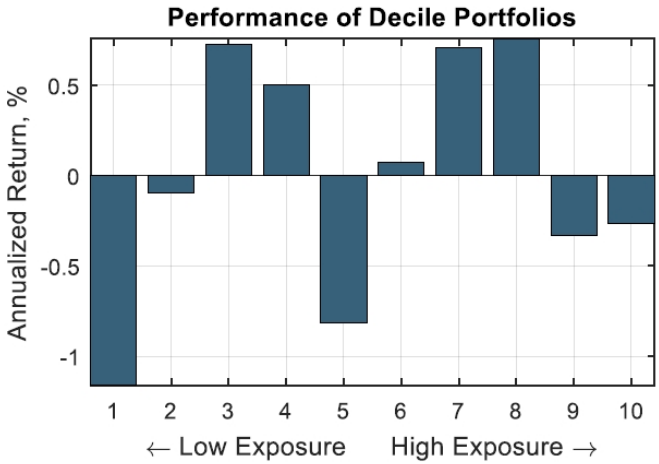
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- **Description:** Describes stock return differences due to various ratios of the company's earnings relative to its price.
- **Motivation:** The Earnings Yield factor is one of the relative valuation multiples popular in industry.
- **Exposure:** A positive exposure indicates a high earnings yield ('cheap' stocks). A negative exposure indicates a low earnings yield ('expensive' stocks).
- **Variable:** Cash Earnings-to-Price / Enterprise Multiple (EBIT to EV) / Earnings-to-Price /Analyst-Predicted Earnings-to-Price



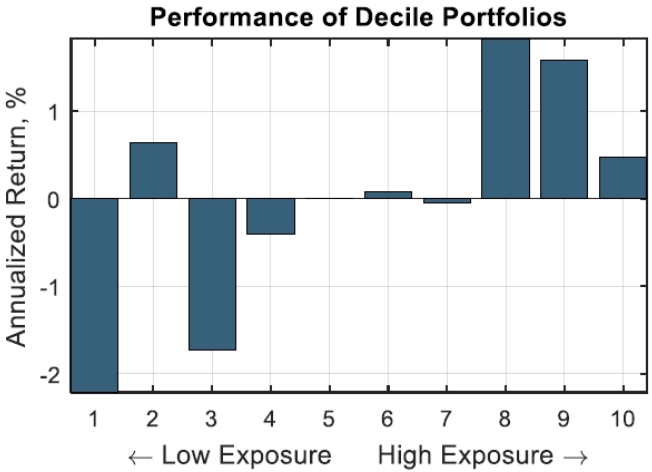
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- **Description:** Measures company growth prospects using historical sales growth and historical and predicted earnings growth.
- **Motivation:** Growth is one of the factors that determine the future cash flow and dividends paid out to investors and, thus, future stock prices.
- **Exposure:** A positive exposure indicates a high growth. A negative exposure indicates a low growth.
- **Variable:** Analyst Predicted Earnings Long-Term Growth / EPS Growth Rate / Sales Per Share Growth Rate



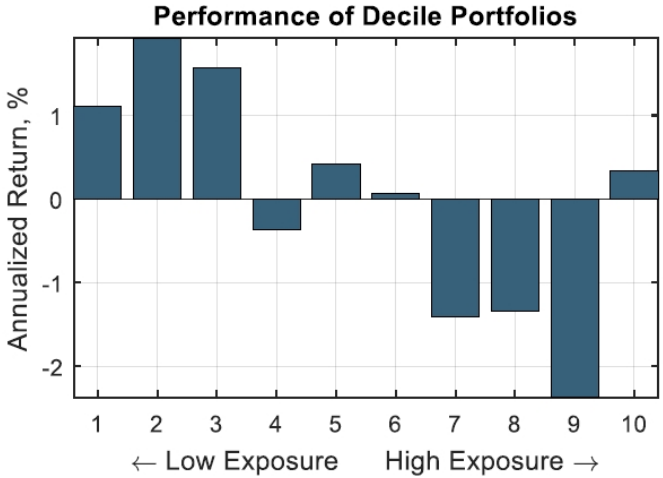
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- **Description:** Measures the tendency of management to over-invest and management views when the stock is over/under priced.
- **Motivation:** Companies with corporate events associated with asset expansion (e.g., public equity offering) tend to experience lower returns than companies with corporate events associated with asset contraction (e.g., share repurchase).
- **Exposure:** A positive exposure indicates low asset and capital expenditure growth. A negative exposure indicates high asset and capital expenditure growth.
- **Variable:** Total Assets Growth Rate / Capital Expenditure Growth / Issuance Growth



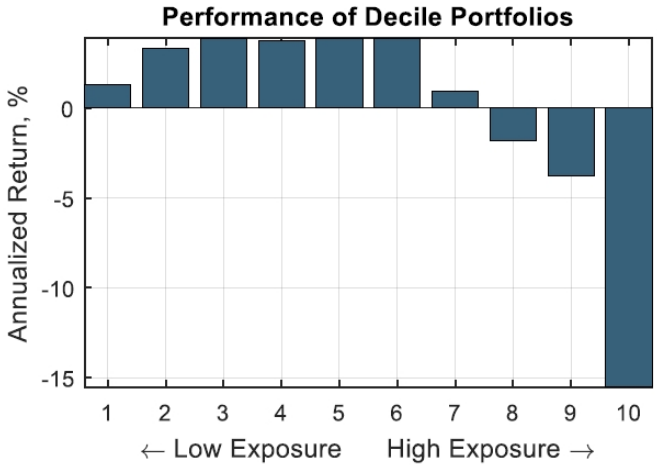
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- **Description:** Captures common variation in stock returns due to differences in the level of company leverage.
- **Motivation:** Highly-leveraged firms are riskier because they cannot change their production easily.
- **Exposure:** A positive exposure indicates a high leverage. A negative exposure indicates a low leverage.
- **Variable:** Debt-to-Assets / Book Leverage / Market Leverage



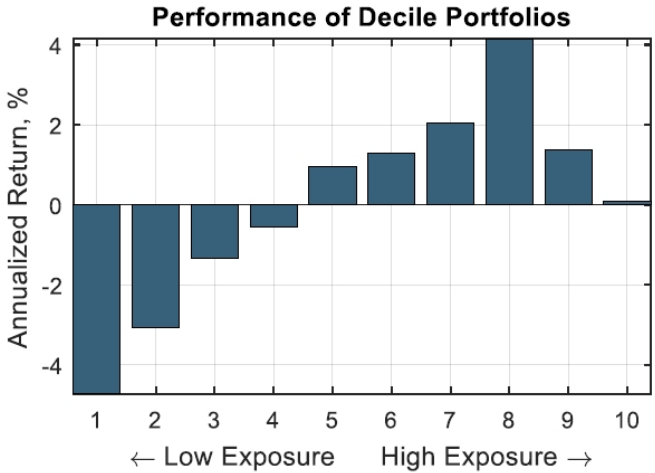
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- **Description:** Captures common variations in stock returns due to differences in the impact of trading on stock returns.
- **Motivation:** The ability of an investor to convert stock holdings into cash determines an illiquidity premium, that is, excess returns that investors require for holding difficult-to-sell illiquid stocks.
- **Exposure:** A positive exposure indicates a high liquidity. A negative exposure indicates a low liquidity.
- **Variable:** Monthly Share Turnover / Quarterly Share Turnover / Annual Share Turnover / Annual Traded Value Ratio



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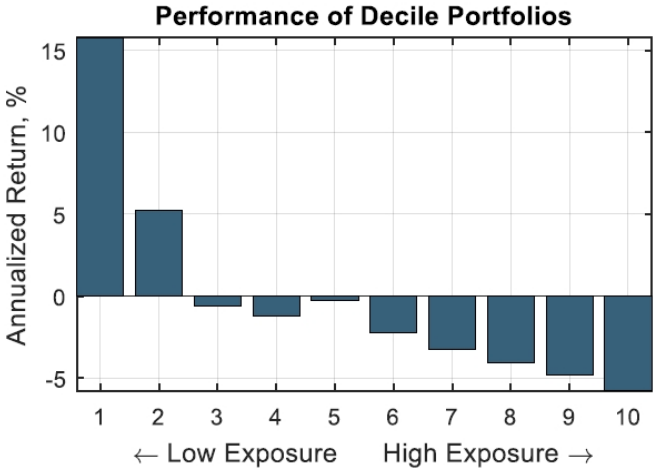
- **Description:** Explains common variation in returns related to a long-term (five years ex. recent thirteen months) stock price behavior.
- **Motivation:** The early evidence for long-term reversal phenomena in stock returns goes back to De Bondt and Thaler (1985).
- **Exposure:** A positive exposure indicates a low long-term momentum. A negative exposure indicates a high long-term momentum.
- **Variable:** Long-Term Relative Strength / Long-Term Historical Alpha



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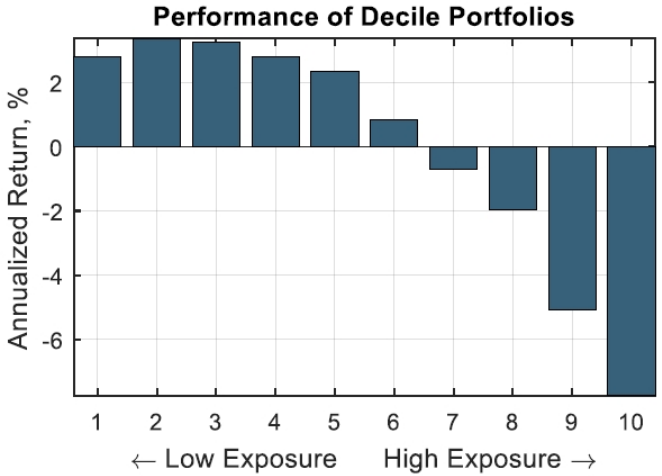
Mid Capitalization (Nonlinear Size)

- **Description:** Captures deviations from linearity in the relationship between returns and the logarithm of market capitalization.
- **Motivation:** A change in expected returns and risk tends to increase more rapidly than implied by a linear model as we move from large-cap to small-cap companies.
- **Exposure:** A positive exposure indicates mid capitalization. A negative exposure indicates the large and small capitalization.
- **Variable:** Regress Size^3 on Size and take the residual.



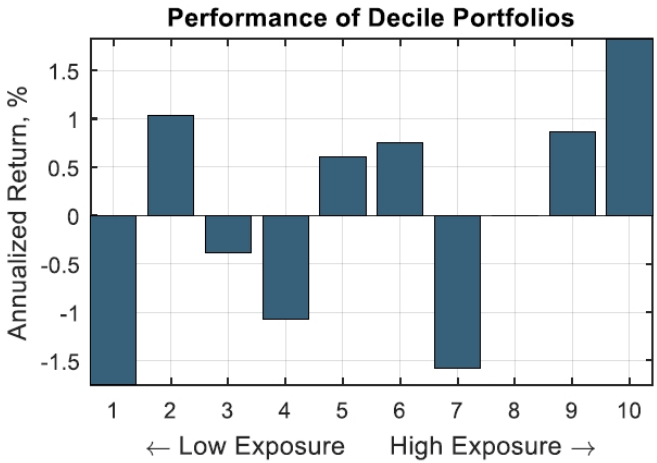
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- **Description:** Explains common variation in stock returns related to recent (twelve months with a lag of one month) stock price behavior.
- **Motivation:** The Momentum factor.
- **Exposure:** A positive exposure indicates a high medium-term momentum. A negative exposure indicates a low medium-term momentum.
- **Variable:** Historical Alpha / Relative Strength 12-month



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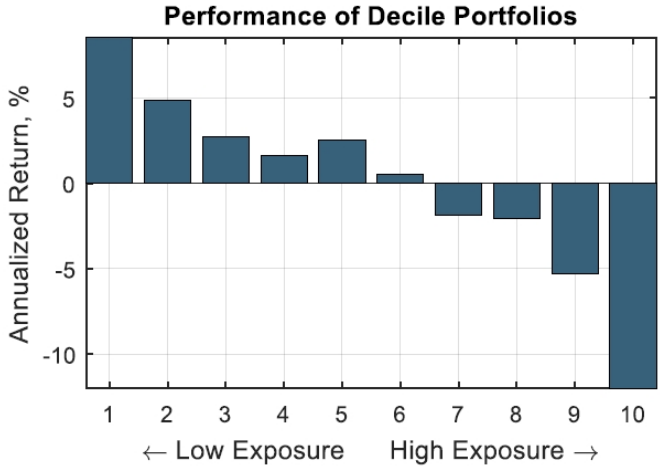
- **Description:** A combination of profitability measures that characterizes efficiency of a firm's operations and total activities.
- **Motivation:** The Dividend Discount Model implies that higher expected future earnings imply a higher expected stock return.
- **Exposure:** A positive exposure indicates a high profitability and operating efficiency. A negative exposure indicates a low profitability and operating efficiency.
- **Variable:** Asset Turnover / Gross Profitability / Gross Profit Margin / Return on Assets



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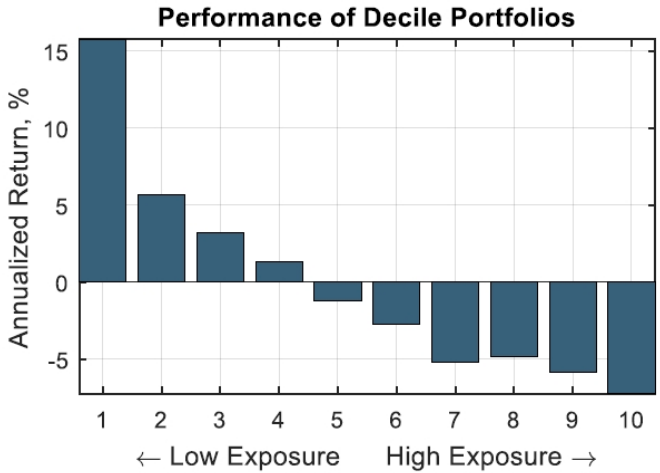
Residual Volatility

- **Description:** Captures relative volatility in stock returns that is not explained by differences in stock sensitivities to market returns.
- **Motivation:** The Idiosyncratic Volatility anomaly.
- **Exposure:** A positive exposure indicates a high residual volatility. A negative exposure indicates a low residual volatility.
- **Variable:** Historical Sigma / Daily Standard Deviation / Cumulative Range



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- **Description:** Captures differences in stock returns and risk due to differences in of the market capitalization of companies.
- **Motivation:** The Size factor.
- **Exposure:** A positive exposure indicates large capitalization. A negative exposure indicates small capitalization.
- **Variable:** Log of Market Capitalization



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Questions?

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